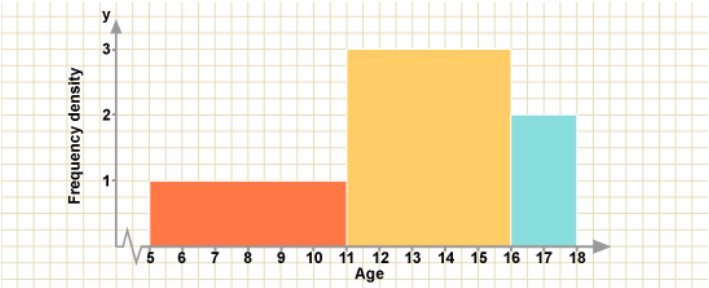
1. Statistics

1.1. Histogram



- A histogram displays frequency of continuous or grouped discrete data using bars
- Bars are joined together and may be of varying width
- Frequency of the data is represented by the area of the bar and not the height. When class intervals are different, area of the bar represents the frequency, not the height
- Frequency density plotted on y-axis, not frequency
- **Class width** = Size of Interval
- **Frequency density** = Height

$$Frequency = Class\ width \times Frequency\ density$$

1.2. Averages

Mean

$\frac{\text{Sum of values}}{\text{number of values}}$

Estimated Mean (Grouped Data)

- Find the midpoint of each group,
- Multiple each midpoint by its frequency,
- Add the results and divide by the total frequency.,

$$\text{Estimated Mean} = \frac{\sum(\text{Midpoint} \times \text{Frequency})}{\text{Total Frequency}}$$

Median

- The middle value - when the data has been written in ascending or descending order
- Odd No. of Values: (n+1)/2 th value
- Example: For 5 values → (5+1)/2 = 3rd Value
- Even No. of Values: (n)/2 and (n/2 +1)th values
- Take average of these two values

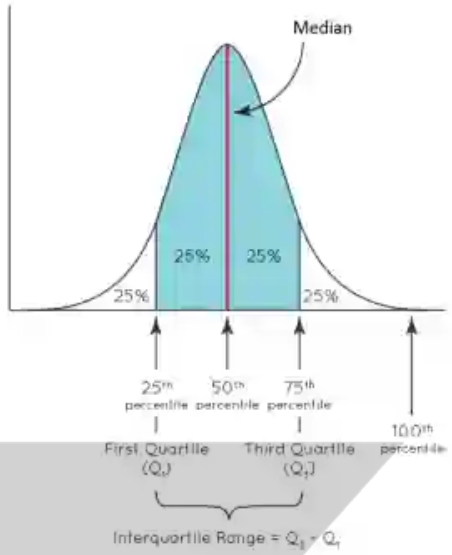
Mode

- Most frequently occurring value

Range

- Range = Highest Value - Lowest Value

- Quartiles



- Upper Quartile (Q3) -

$$\frac{3}{4}(n+1)$$

- Lower Quartile (Q1) -

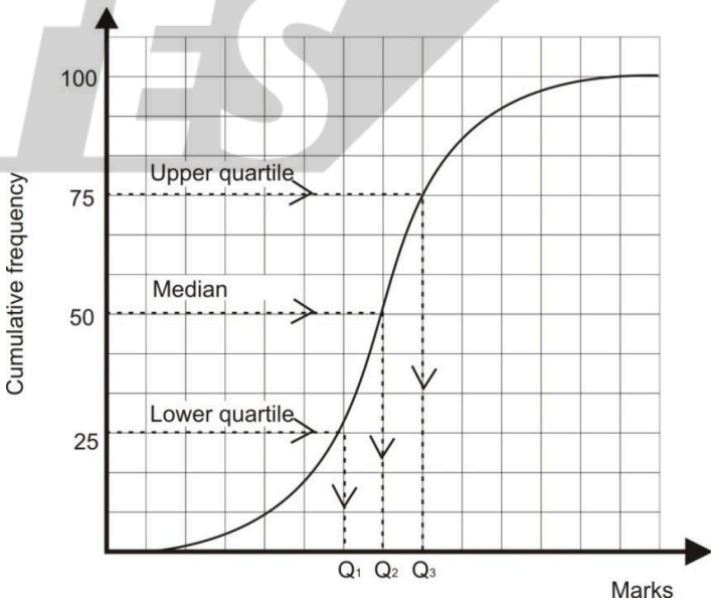
$$\frac{1}{4}(n+1)$$

- n = total number of data in a set

Modal Class Interval - Class interval with the highest frequency

1.3. Understanding Cumulative Frequency

- Cumulative frequency is the total frequency up to a given point
- Inter-quartile range $= upper\ quartile - lower\ quartile$
- Interpreting Cumulative Frequency Diagrams



- The cumulative frequency graph must always follow a sigmoid curve shape

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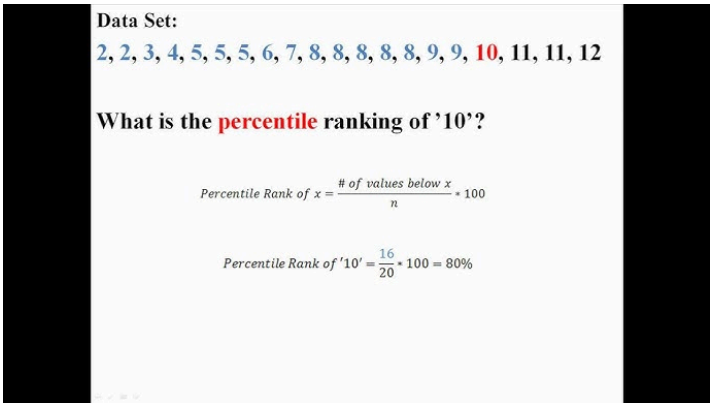
- Percentiles



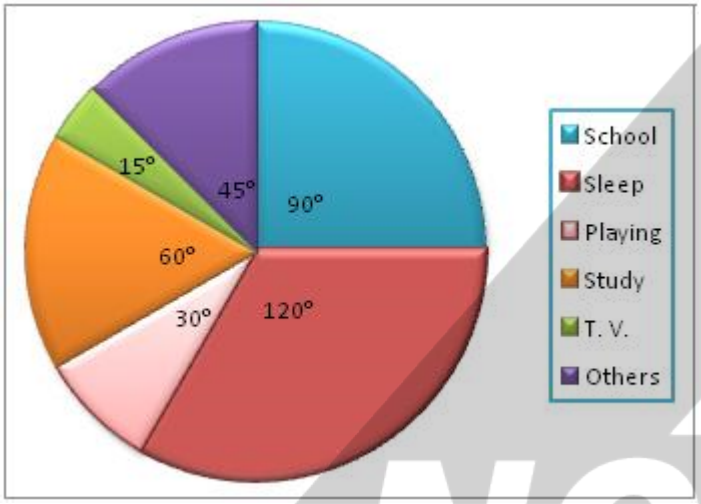
1.4. Pie Charts

- **Definition -**

  Sectors represent data, and these sectors form a circle.

- **Angle of a sector:**

$$\theta = \frac{\text{Number of an item}}{\text{Total number of items}} \times 360^\circ$$



- Sum of angles in a pie chart is 360°
- Calculate the angle for each sector representing a category, then use a protractor to draw the sectors within a circle

1.5. Stem and Leaf diagrams

Definition

- Stem-and-Leaf diagram is a quick way of summarizing a range of data.
- There is a column known as the stem, which contains unique elements of data formed by removing last digits of the data.
- Keys are used in this diagram

Structure

stem	leaf
0	1, 1, 2, 2, 3, 4, 4, 4, 4, 5, 8
1	0, 0, 0, 1, 1, 3, 7, 9
2	5, 5, 7, 7, 8, 8, 9, 9
3	0, 1, 1, 1, 2, 2, 2, 4, 5
4	0, 4, 8, 9
5	2, 6, 7, 7, 8
6	3, 6

Key: 6|3 = 63 years old

- A key or legend is always included to show how to read the diagram.

How to find the median of a Stem leaf diagram

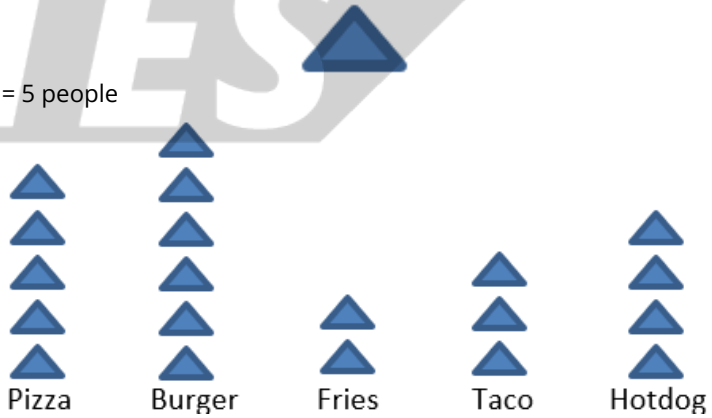
- First ensure the data is ordered from smallest to largest .
- Locate the middle value.
- If there's an odd number of data points, the median is the single middle value.
- If there's an even number of data points, the median is the average of the two middle values

1.6. Pictograms

- Data is represented in pictures
- A key is given to represent the value of a picture.
- There are no axes
- Frequency is represented by symbols
- A key shows value of 1 symbol

E.g.

= 5 people



Favorite Fast Food of 100 Children

1.7. Scatter Diagrams

- Displays the correlation between two sets of data
- May have positive, negative or no correlation
- Scatter graphs are used to plot pairs of data

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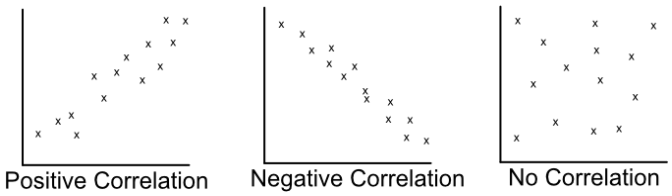
- The vertical and horizontal axes represent the two quantities being measured
- Points are plotted as crosses, ×
- The general shape formed by the points shows the type of correlation

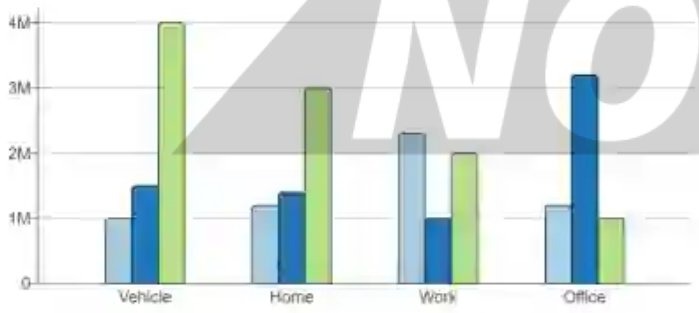


- **Correlation**
- Positive: Indicates a relationship between two variables where they tend to move in the same direction
- Negative: An inverse relationship between two variables, meaning as one variable increases, the other decreases
- No Correlation: No derivable relationship between the two sets of data being plotted

- **Line of Best Fit**
- Use a transparent ruler to visually estimate the trend of the data points
- Draw a straight line that represents that trend.
- Aim for roughly the same number of data points above and below the line

1.8. Bar Charts

- Used to represent discrete data (data in separate categories).
- Each bar represents a category and its frequency (or value).
- Bars must be of equal width and equally spaced



Key Features

- Horizontal Axis shows categories
- Vertical axis shows frequency or measured value
- Height of Bar = Frequency
- Used when data is categorical and discrete

Reading a Bar Chart

- Look at the height of each bar
- Match it with the scale on the y-axis
- Compare bars to see the highest, lowest, or differences

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